

CLAUDE IN PHARMA MANUFACTURING

A plant that learns from every batch

The answers are already in your SOPs, batch records, deviations and training data. Claude reads them together, so people can act on them sooner.

TODAY

Reactive

Something goes wrong, then the investigation starts. People pull evidence by hand from quality, training and document systems.

NEXT

Proactive

Claude scans the same records on a schedule and flags the pattern before it becomes a deviation or a CAPA.

LATER

Connected

Agents work with equipment, cameras and digital batch records, with a person approving every action that matters.

Read-only to start. Every output is a draft for a named approver.

Where Claude fits across the plant

Quality

Deviation investigation, root cause review, CAPA drafting and batch record review, with every claim cited to its source.

In the demo

Training

SOP changes turned into targeted training, practice built from real deviations, and checks that the training worked.

Tech transfer and CMC

Process knowledge carried from development to the plant and between sites, without rebuilding it from paper each time.

New facilities

Commissioning, qualification and validation documents drafted and tracked, with schedule risks surfaced early.

Daily operations

Shift handovers, line status and next best actions pulled into one view for supervisors.

Knowledge

One place to ask what the procedures say, where they disagree, and what experienced staff know that isn't written down.

Same pattern in every box: connect the systems, write the method down as a skill, keep a person on the decision

The answer sits across four systems

System	Claude reads
Quality management	The deviation record and past deviations
Lab information	Test results and calibration records
Batch record	The batch record and raw material genealogy
Equipment historian	Equipment sensor data over time

The decision stays human

Claude reads from all four and writes one thing: a draft, routed to QA.

A qualified reviewer checks the sources and signs. CAPAs are created only after approval.

No single system holds the whole picture. In the demo these are mock systems with fictional data.

Signals from the industry

BRISTOL MYERS SQUIBB

Claude across the enterprise, manufacturing included

An agreement to bring Claude to more than 30,000 employees. BMS will evaluate it for root cause investigation, CAPA documentation and data-driven batch release.

BMS press release, May 20, 2026

MODEL HARDWARE STANDARD

A shared way for agents to operate equipment

A research preview, open to a first group of labs and advanced manufacturers. Carnegie Mellon built its drivers and orchestration in about eight hours, against several weeks for a vendor setup.

Anthropic, Aug 27, 2026

Both are public announcements. BMS describes plans to evaluate, not results.

WHERE IT GOES

Closer to the floor, ahead of the deviation

BE MORE PROACTIVE

Runs on the systems you have today

Drafts the deviation

An operator describes the event. Claude drafts the record with SOP and history linked. QA classifies.

Alerts the supervisor

Training gaps and deviation history point to a risk. The supervisor gets the evidence and a plan.

Recommends the fix early

When near misses cluster on an unclear SOP step, Claude drafts the change before there is a CAPA.

CONNECT TO THE PHYSICAL WORLD

Adds cameras, digital batch records and equipment

Cameras confirm the sequence

Vision models check each intervention, not only the ones a supervisor happens to watch.

Trends from the floor

Digital batch records and logbooks show who did what and when, so alerts get sharper.

Agents that run equipment

The Model Hardware Standard, now in research preview, is the path to agents operating devices safely.

How to start: one workflow, read-only connections, and a named approver on every output.

Longer-term direction, not a product commitment. Framed as closing gaps, never as scoring operators.

BEFORE THE DEMO

The scenario: a pH excursion on a drug substance batch

WHAT HAPPENED

A pH reading of 7.82 on a bioreactor batch, above the 7.5 limit. It is logged as a major deviation and the batch goes on hold.

WHAT CLAUDE PULLS

Evidence from four systems at once: quality, lab, batch record and equipment historian.

WHAT COMES OUT

A draft investigation: ranked root causes with sources, recommended CAPAs and a batch disposition, routed to QA.

Watch for: which system holds the clue, and what the deviation history shows.

 **Sentinel QMS**
Synced 2 min ago LIVE

Open Deviations 6

- DEV-2026-0847
pH Excursion – Batch RX-4521
Just now MAJOR
- DEV-2026-0842
Particulate OOS – Lot PM-1192
2h ago CRITICAL
- DEV-2026-0838
Fill Volume Drift – Line 3
Yesterday MINOR
- DEV-2026-0831
Temp Excursion – Cold Room B
Mar 28 MAJOR
- DEV-2026-0825
Raw Material ID Fail – RM-0456
Mar 27 MINOR

Root-Cause Investigation

DEV-2026-0847 · pH Excursion · Batch RX-4521

DRAFT MAJOR

🌟 CO₂ pressure anomaly found in Historian ...

● Observation

pH reading of **7.82** at T+72h, exceeding spec limit of 7.5. Returned to **7.31** within 2h after CO₂ overlay pressure restored.

● Likely Root Cause(s)

Three hypotheses evaluated against the evidence. Ranked by likelihood — final determination pending QA review.

MOST LIKELY CO₂ overlay regulator valve degradation (R-204)

High confidence

CO₂ pressure dropped 0.15 → 0.04 bar at T+68h, 4h before the excursion. Pattern matches 3 prior deviations on the same unit.

[Chronos: CO₂ pressure trend](#) [3 prior deviations](#) [R-204 maintenance log](#)

5M: Machine

POSSIBLE In-line pH probe calibration drift

Moderate — not excluded

Probe last calibrated 11 days prior (within 14-day SOP limit). Cannot be fully excluded without re-calibration check; recommended as a verification action.

[Helix: calibration record](#) [SOP-BIO-039](#)

5M: Method

UNLIKELY Raw material / buffer variability

Low — evidence contradicts

All raw materials QC-released; buffer lot used in 6 other batches with no excursions. Genealogy shows no correlation.

[Atlas: batch genealogy](#) [Buffer lot CoA](#)

5M: Material

Recurrence history — primary hypothesis

DEVIATION	DATE	ASSIGNED CAUSE	GAP
DEV-2024-1102	Jan 2025	CO ₂ valve wear	—
DEV-2025-0234	Jul 2025	CO ₂ valve wear	6mo
DEV-2025-0891	Nov 2025	CO ₂ valve wear	4mo
DEV-2026-0847	Mar 2026	Under review	3mo

System Data 4/4

Sentinel QMS LIVE
Deviation DEV-2026-0847
Classification Major

Helix LIMS LIVE
Result 7.82 at T+72h
Spec 6.5–7.5

Atlas ERP (Batch Record) LIVE
Process Order RX-4521 · R-204
Batch Value \$1.4M

Chronos Historian LIVE
CO₂ Pressure 0.04 bar at T+68h
Setpoint 0.15 bar

🌟 Agent Activity Working

- 🔍

Sentinel QMS
Deviation record retrieved
- 📊

Helix LIMS
14 pH readings — excursion at T+72h
- 📋

Atlas ERP
Batch record + raw material genealogy
- 🕒

Chronos Historian
CO₂ sensor data — pressure drop at T+68h
- 🔍

Analysis
Causal pattern identified
- 📊

Pattern Match
3 prior deviations found
- 🔍

Trend Alert
Accelerating failure pattern